# Electrical Energy and Power

(1) A factory is served by 220 V power supply line. In a circuit protected by a fuse marked 10A, find the maximum no of 100W bulbs in parallel that can be turned on? (22)

(2) A heating coil is labelled 100W – 220V. The coil is cut in two halves and the two pieces are joined in parallel to the same source. Find total power dissipated now? (400W)

(3) A 220V – 1000 watt bulb is connected across a 110-volt mains supply. Find power consumed. (250W)

(4) A battery is first connected to an external resistance 'R₁' and then connected to other ext resistance 'R₂' (removing 'R₁'). For what value of internal resistance of the battery heat liberated per second in both ext resistance will be the same ($r = \sqrt{R_1 R_2}$)

(5) The burner of an electric stove is made up of three equal resistances. If they are connected in parallel, water in a tea kettle begins to boil in 6 min. When will the same mass of water in the tea kettle begin to boil if the resistors are connected in series? (54 min)

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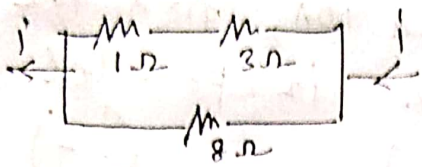
6. Power dissipated across the 8 Ω resistor (in fig) is 2W. Find the power dissipated across the 3Ω resistor [3W]

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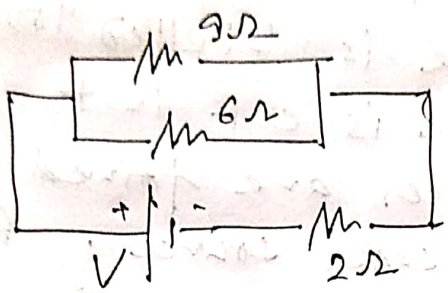
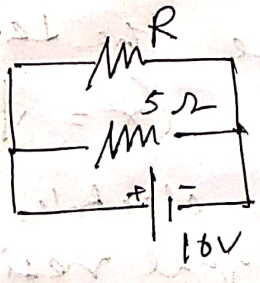



Power dissipated in 9Ω resistor is 36W. Find p.d across 2Ω resistor. (10V)

8. If voltage across a bulb rated 220V-100W drops by 2.5% of its rated value, find the % of the rated value by which the power would decrease? (5%)

9. In India supply voltage for domestic use is 220V & it is 110V in USA. If resistance of a 60W bulb is 'R' in India, then what would be its resistance in USA. (R/4)

10.




Total power dissipated by this (fig) circuit is 30W. Find R.

11. Through a supply line of resistance 0.5Ω, Power of 15kW is transmitted at 250V p.d. Find efficiency of the supply. (88%)

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(12) In a ~~school~~ house there are 6, 40W and 2, 100 W bulbs, 4, 40 W electric fans and a 1000 W electric heater. Every bulb is on for 5 hr, every fan runs for 15 hr and the heater runs for 2 hr every day. What will be electric bill for April. [cost for 1 BOT unit is Rs 1.5/-]

[Rs 297/-] If supply voltage is 200 volt then, what is the value of minm. fuse connected in the house [8A].

(13) - What resistance should be connected in series to a bulb 60V - 120W, so that it glows brighter (i.e dissipates same power) when connected in a 120 volt d.c supply.

[80Ω]

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(14) An electric heater has a resistance of 12Ω and is operated from 220 V power line. No heat escapes from it, how much time is required to raise the temperature of 40 kg of water from 15°C to 80°C ?

(2730 sec Approx)

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